8a	The value of V when I = 0 remains unchanged as intensity is varied. This value is the stopping potential.	B1 B1
8bi	$Q = It$ and $ne = It$ Hence $n = It / e = (1.2 \times 10^{-7} \times 5.0) / 1.6 \times 10^{-19}$ $n = 3.75 \times 10^{12} \approx 3.8 \times 10^{12}$	 C1 A1
8bii	Using Einstein's photoelectric equation : $hf = \Phi + KE_{max}$ $KE_{max} = hf - \Phi$ $= 6.63 \times 10^{-34} \times 7.0 \times 10^{14} - 2.2 \times 1.6 \times 10^{-19}$ $= 1.12 \times 10^{-19} \approx 1.1 \times 10^{-19}$ J	 C1 A1
8biii1	The energy of a photon is given by hc / λ, therefore the energy of each photon remains the same/constant (since λ is constant). For the same metal the work function remains the same, ,hence (from the photoelectric equation) the maximum kinetic energy remains the same/constant.	M1 M1 A1
8biii2	There are twice as many photons reaching the surface, hence the current in the photocell doubled	M1 A1
8biv	Zinc has work function larger than the photon energy, Hence no electrons ejected, so no photocurrent.	B1 B1
8ci	Fast-moving electrons pass through thin carbon/graphite Concentric diffraction rings are observed on the screen	B1 B1
8cii	From $\lambda = h/p = h/mv$ $= 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 2.5 \times 10^{6})$ $= 2.91 \times 10^{-10} \approx 2.9 \times 10^{-10}$ m	 A1
8ciii	They are suitable as their wavelength is comparable to the atomic spacing.	B1
8civ	Neutrons have no charge (advantage or disadvantage) Advantage: Neutrons experience no electrical forces (when interacting with atoms) Disadvantage: Neutrons cannot be accelerated using an accelerating potential hence difficult to control their speeds. Neutrons production (require a nuclear reactor, and thus) is much more difficult than electron production	B1 B1 B1